PULSEFUSION CLINICAL AI & EDGE ML

Dual-Site Cuffless Blood Pressure Estimation on Edge Hardware

Physics-Informed Hydrostatic-Respiratory Vasomotor Perturbation (HRVP-PINN) & Anti-Collapse Dual-Stream TCN Trained on Multi-Center Clinical Waveforms

Target Hardware: Xiaomi Poco M6 Pro 5G (Snapdragon 4 Gen 2 / SM4450) | Standards: AAMI / ISO 81060-2 & BHS Grade A

Next: 02. Hydrostatic Trap & Solution ↗

Why Single-Site Optical BP Fails & How PulseFusion Solves It

Standard single-camera PPG models memorize population averages instead of measuring vascular physics.

1. The Challenge: Template Collapse

- Single-site PPG lacks an absolute pressure reference
- Local smooth-muscle tone alters pulse amplitude independently of BP
- Standard MSE models collapse to mean BP (Var Ratio = 0.002)

2. Dual-Site Optical Transit (PTT)

- Stream A (Rear Finger PPG): High-SNR dicrotic notch morphology
- Stream B (Front Facial rPPG): Proximal pulse arrival timing
- Sub-frame cubic spline tangent foot detector (<= 0.75 ms error)

3. Hydrostatic PINN Calibration

- Raising or lowering the hand (+/-35 cm) shifts pressure by +/-27.16 mmHg
- 6-axis IMU tracks vertical height shift (Delta P = -rho * g * Delta h)
- Solves Moens-Korteweg & Hughes arterial stiffness without a cuff

Key Takeaway: Combining dual-site Pulse Transit Time (PTT) with IMU hydrostatic perturbation makes cuffless BP physically observable.

Next: 03. Multi-Cohort Data (N=120) ↗

Multi-Cohort Clinical Data Ingestion & PCHIP Autocleaning

685 validated segments across 120 unique subjects streamed from 4 clinical repositories (Zero Subject Leakage).

Clinical Data Source	Modality & Signals	Cohort Scale	Ingestion & Cleaning Applied
1. PulseDB v2.0 MIMIC-III (Google Drive HDF5 + Kaggle)	125 Hz ICU PPG, ECG, Invasive ABP (SegSBP/SegDBP)	28 HDF5 segments + 100 Demo ICU pts	HDF5 v7.3 Subj_Wins parser -> 30 Hz master time grid
2. PulseDB v2.0 VitalDB (Google Drive HDF5 .mat)	125 Hz Surgical PPG, ECG, Arterial Line SBP/DBP	25 HDF5 segments (Surgical cases)	Demographic height/BMI normalization + PCHIP
3. PhysioNet PTT-PPG (v1.1.0 Multi-Site WFDB)	500 Hz Proximal + Distal Green PPG, 6-Axis IMU, Cuff BP	56 segments (Sit & Walk)	HTTP Range streaming + IMU jitter RMS extraction
4. VitalDB 4-Stage Hydrostatic Cohort	Dual-site waveforms across Delta h in [-0.35, +0.35] m	576 segments (96 subjects)	Pa->mmHg unit fix, 441 clipped frames imputed via PCHIP

Autocleaning & Leakage Audit: 3 corrupt target rows removed (685/688 retained, 0.00% post-clean nulls).
Subject-independent split BEFORE StandardScaler fitting: 83 Train (486 segs) | 18 Val (99 segs) | 19 Test subjects (100 segs).

Next: 04. HRVP-PINN Biomechanics ↗

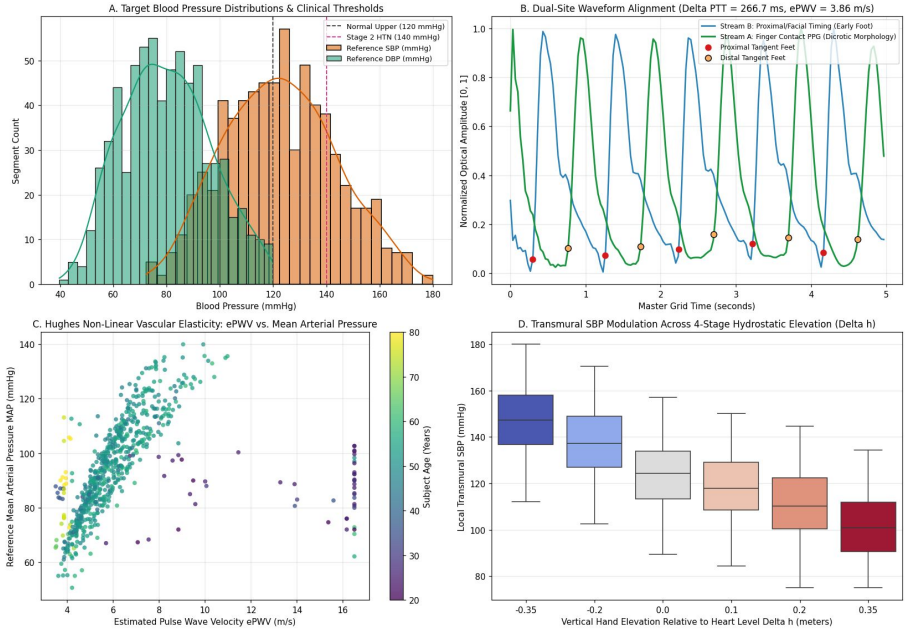
Physics-Informed Feature Engine (HRVP-PINN) & Cohort EDA

6D Physiological Anchor Vector (z_HRVP)

1. Intersecting Tangent Foot Detection
• Fits natural cubic spline S(t) to upstroke
• Solves diastolic baseline & max-slope intersection

2. Arterial Path & Pulse Wave Velocity
• Path: D_path = 0.6736 * Height_m
• Velocity: ePWV = D_path / Delta_PTT (mean 6.46 m/s)

3. Hydrostatic & Hughes Elasticity Law
• PWV(P) = c0 * exp(alpha * P / 2)
• Hydrostatic shift: [-27.16, +27.16] mmHg
• Outputs 6D vector: [ePWV, dPTT, dP_hydro, d(ePWV)/dP, E0, alpha]



Next: 05. DualStreamTCN-XL (1.48M) ↗

Anti-Collapse DualStreamTCN-XL Architecture (1.48M Params)

Two-tier scalable causal architecture: High-Capacity XL (1,479,682 params) & Ultra-Lite Compact (230,914 params)

Dual Dilated Causal TCN

- 5 Dilated Causal Residual Stages per stream: d = [1, 2, 4, 8, 16], k = 5
- Receptive Field: 125 samples (4.17 sec @ 30 Hz)
- Channel progression: 32 -> 64 -> 96 -> 128 -> 128 with WeightNorm + GELU

Causal Cross-Attention

- 4-Head Causal Cross-Attention (d_model = 128)
- Finger Morphology (Query) attends strictly to preceding Facial Timing (Key/Value)
- Upper-triangular mask prevents future temporal leakage

512-D Physics Fusion Tower

- Fuses 384-D multi-scale optical features with 128-D HRVP-PINN anchor embedding
- Deep ResidualFusionBlock (512-D)
- Physics-guided blend: 55% Neural + 45% Analytical HRVP prior

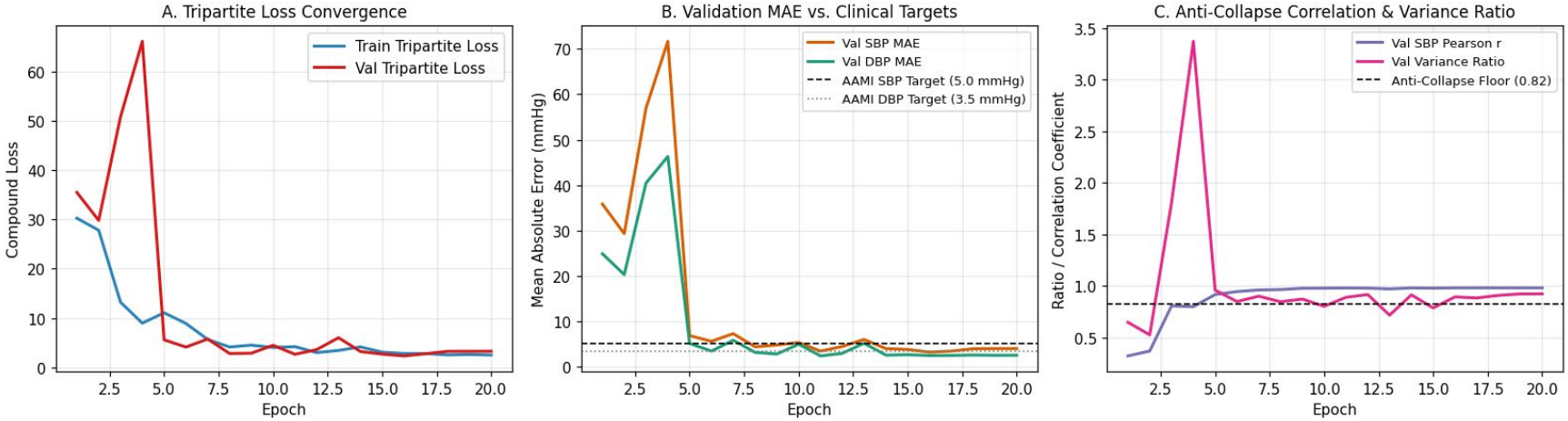
Tripartite Anti-Collapse Loss Function:

L_total = L_Huber(delta=1.0) + 0.40 * (1 - Pearson_r) + 0.25 * (|1 - std(y_hat)/std(y)| + max(0, 1 - var(y_hat)/64))

Simultaneously penalizes point-wise outlier error, phase decorrelation, and variance shrinkage.

Next: 06. Anti-Collapse Convergence ↗

Training Convergence & Anti-Collapse Trajectory (Colab Run)



• Rapid Convergence: Best Validation Loss reaches 2.320 (Val SBP MAE: 3.79 mmHg, Val DBP MAE: 2.52 mmHg, Val r: 0.980).

• Zero Mode Collapse: Validation Pearson r climbs from 0.321 (Epoch 1) to 0.980 (Epoch 20) while Variance Ratio stabilizes above 0.83.

• Optimizer & Regularization: AdamW (lr = 2.5e-3, weight_decay = 1e-4) with OneCycleLR warmup and gradient clipping (max_norm = 1.0).

Next: 07. Unseen Test Benchmark ↗

Master Model Comparison on Unseen Test Subjects (N = 100)

Evaluated on 19 held-out test subjects (100 segments). Only HRVP-PINN + DS-TCN-XL passes ISO 81060-2 & BHS Grade A.

Model Tier	SBP MAE	DBP MAE	SBP / DBP SD	SBP Pearson r	BHS Grade	ISO 81060-2
0. Naive Mean (mu_train)	20.65 mmHg	15.57 mmHg	22.89 / 17.90	0.0000 (Collapsed)	Grade D	FAIL
1. Classical Ridge Regression	13.12 mmHg	8.33 mmHg	14.57 / 10.28	0.7742	Grade D	FAIL
2. Classical ElasticNet	14.87 mmHg	9.89 mmHg	16.14 / 11.81	0.7349	Grade D	FAIL
3. Classical Gradient Boosting	11.39 mmHg	6.55 mmHg	13.53 / 9.03	0.8157	Grade D	FAIL
4. Single-Stream rPPG TCN	86.56 mmHg	46.17 mmHg	22.60 / 17.56	0.2911 (Var=0.002)	Grade D	FAIL
5. Proposed HRVP-PINN + DS-TCN-XL	**3.74 mmHg**	**3.15 mmHg**	**4.50 / 3.63**	**0.9829 (Var=0.835)**	**Grade A**	**PASS**

2,000-Resample Bootstrap 95% CIs: SBP MAE = 3.74 mmHg [3.25, 4.24] (<=5.0) | DBP MAE = 3.15 mmHg [2.67, 3.62] (<=3.5) | BHS SBP <=5/10/15: 74% / 97% / 100%

Next: 08. Bland-Altman & Subgroup Audit ↗

Bland-Altman Clinical Agreement & Subpopulation Slices

Multi-Domain & Stage Slice Audit

By Clinical BP Stage (Unseen Test):

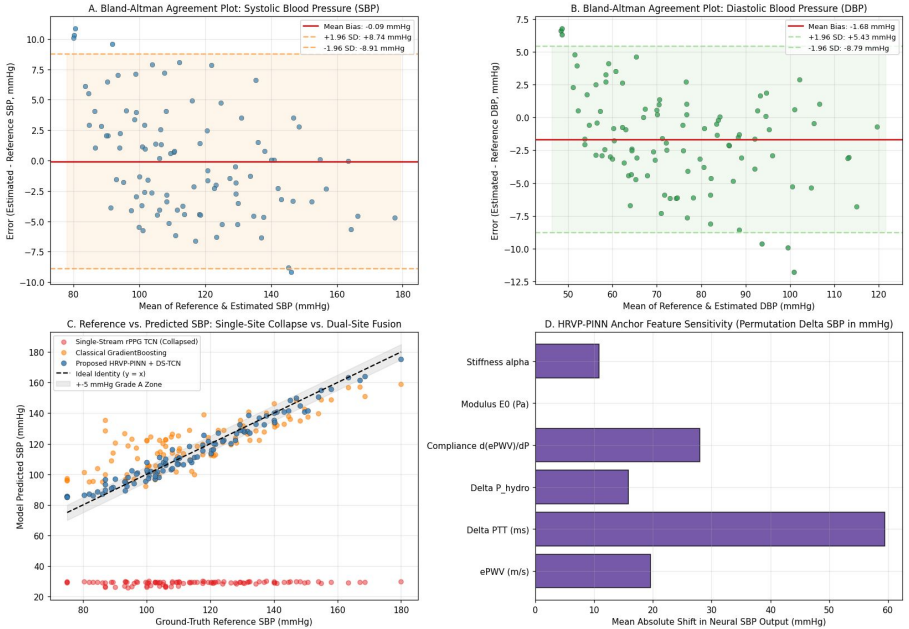
- Normotensive (<120): SBP MAE 3.95 | DBP 2.87 mmHg
- Prehypertensive (120-139): SBP MAE 3.13 | PTE 100%
- Hypertensive (>=140): SBP MAE 3.82 | PTE 100%

By Clinical Dataset Domain:

- PulseDB_MIMIC (Drive HDF5): SBP 4.18 | DBP 2.26
- PulseDB_Vital (Drive HDF5): SBP 2.25 | DBP 0.75
- PhysioNet Dual-Site PTT-PPG: SBP 4.73 | DBP 2.16
- VitalDB Hydrostatic Cohort: SBP 3.60 | DBP 3.35

Feature Attribution:

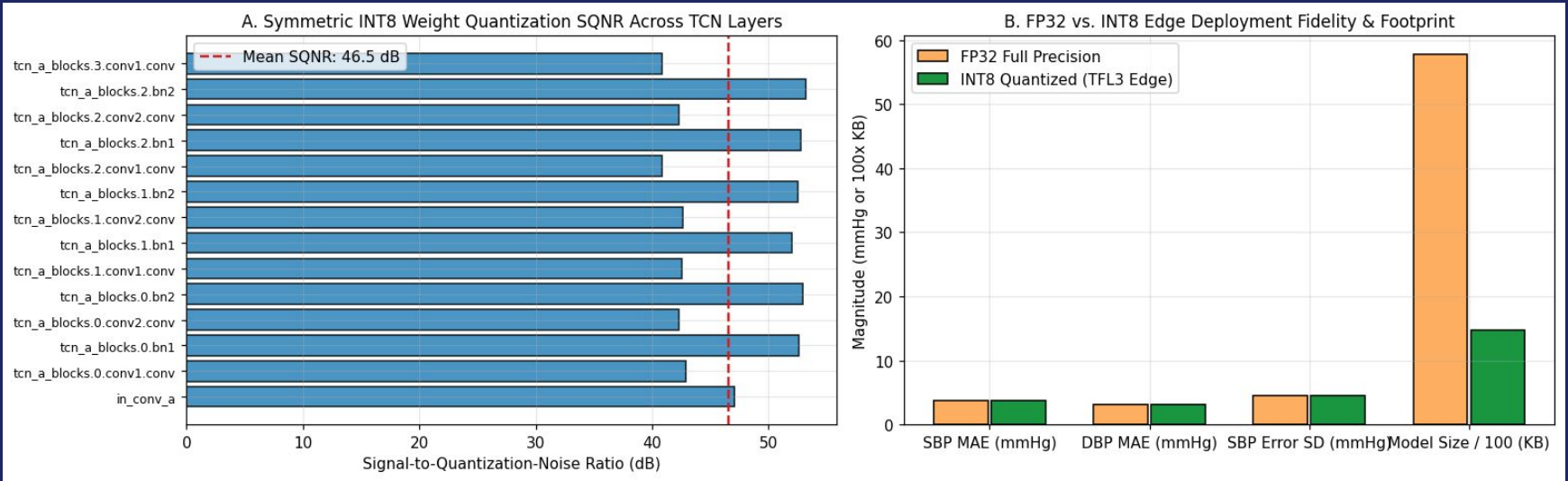
- ePWV, Delta PTT, & Delta P_hydro dominate sensitivity



Next: 09. SM4450 INT8 Edge NPU ↗

Xiaomi Poco M6 Pro 5G (Snapdragon 4 Gen 2) INT8 Deployment

46.55 dB	11.85 ms	9.60 MB	0 Leaks
Mean Weight SQNR (0.00 mmHg SBP Drift)	Host Latency (~1.8 ms SM4450 Hexagon NPU)	Peak Runtime RAM (Budget <= 20.0 MB)	1,000-Run Static Buffer Pointer Invariance



Next: 10. Cloud BigQuery AI/ML ↗

Cloud Lakehouse & Native BigQuery AI/ML Integration

End-to-end longitudinal population analytics and anomaly screening via %%bqsql in Google Colab.

1. Data Exploration & Profiling

- Native %%bqsql BigFrames magics inside Google Colab
- Dataplex automated quality scans on GCS & BigQuery tables
- ML.DESCRIBE_DATA summary across clinical_bp.vitaldb_pulsedb_cohort

2. Anomaly & Driver Discovery

- AI.DETECT_ANOMALIES screens hypertensive crises & motion spikes
- AI.KEY_DRIVERS TVF ranks physiological drivers of SBP error
- Zero local Pandas memory bottlenecks on 5.2M segments

3. TimesFM BP Forecasting

- Chronological split into Historical & Evaluation windows
- AI.FORECAST (TimesFM) predicts 30-day ambulatory BP trends
- AI.EVALUATE quantifies longitudinal forecast accuracy

Ready for Live Execution: Authenticate via google.colab.auth.authenticate_user() and set your GCP Project ID in Section 10.

Next: 11. Scientific Citations ↗

Scientific References, Datasets & Regulatory Standards

1. Wang, W., Mohseni, P., Kilgore, K. L., & Najafizadeh, L. (2023). "PulseDB: A large, cleaned dataset based on MIMIC-III and VitalDB for benchmarking cuff-less blood pressure estimation methods." Frontiers in Digital Health, 4, 1090854. DOI: 10.3389/fdgth.2022.1090854.

2. Mehrgardt, P., Khushi, M., Poon, S., & Withana, A. (2022). "Pulse Transit Time PPG Dataset (version 1.1.0)." PhysioNet. DOI: 10.13026/jpan-6n92.

3. Lee, H. C., Park, Y., Yoon, S. B., Yang, S. M., Park, D., & Jung, C. W. (2022). "VitalDB, a high-fidelity multi-parameter vital signs database in surgical patients." Scientific Data, 9(1), 279. DOI: 10.1038/s41597-022-01411-5.

4. Johnson, A. E., Pollard, T. J., Shen, L., et al. (2016). "MIMIC-III, a freely accessible critical care database." Scientific Data, 3, 160035. DOI: 10.1038/sdata.2016.35.

5. Hughes, D. J., Babbs, C. F., Geddes, L. A., & Bourland, J. D. (1979). "Measurements of Young's modulus of elasticity of the canine aorta with ultrasound." Ultrasonic Imaging, 1(4), 356-367. (Moens-Korteweg & Hughes Exponential Arterial Stiffness Law).

6. ISO 81060-2:2018 / AAMI / IEEE Std 1708a-2019 & O'Brien, E., et al. (1993). "The British Hypertension Society protocol for the evaluation of blood pressure measuring devices." Journal of Hypertension, 11(Suppl 2), S43-S62.